

Physics Final

Unit 4

1.1 $2 \text{ rpm} = \frac{60 \text{ s}}{1.33 \text{ s/rev}} = \boxed{\text{A.) } 45 \text{ rpm}} \rightarrow \boxed{4.7 \text{ radians sec}^{-1}}$

1.2 $\boxed{\text{B.) } 1.1 \text{ ms}^{-1}}$

1.3 $\omega = 10 \text{ } 10/10 = 100$

$\boxed{\text{D.) } \omega \rightarrow 10\omega}$

1.4 $200 \text{ rpm}, r = 12 \text{ cm}, m = 10 \text{ g}$

$\omega = \frac{200 \cdot 2\pi}{60} = 20.94 \text{ rad/sec} \rightarrow a_c = r \cdot \omega^2 = 0.12 (20.94)^2 = 52.53 \text{ m/s}^2$

$F_c = m \cdot a_c = 0.01 (52.53) \rightarrow 0.53 \text{ N}$

$\boxed{\text{B.) } = 0.53 \text{ N}}$

1.5 $v = r \cdot \omega \rightarrow 1.4 (30) = 42 \text{ m/s}$

$\boxed{\text{C.) } 42 \text{ m/s}}$

Unit 5

2.1 $W = \vec{F} \cdot \vec{x} = (40 \hat{i} \cdot 5 \hat{i} + \hat{j} \cdot (-30 \hat{j})) = 200 \text{ N} \cdot \text{m}$

angle $= \sqrt{40^2 + 30^2} = 50$ $\cos \theta = \frac{200}{50 \cdot 5} = 0.8 \rightarrow \cos^{-1}(0.8) = 37^\circ$

$\boxed{\text{D.) } 200 \text{ N} \cdot \text{m}, 37^\circ}$

2.2 a.) $W = F \cdot d \rightarrow (1000) (10) = \boxed{1000 \text{ J}}$

b.) $KE_i = \frac{1}{2} (m) (v_i^2) = \frac{1}{2} (15 \text{ kg}) (10)^2 = \boxed{0 \text{ J}}$

c.) $KE = \frac{1}{2} (m) (v_f^2)$

$v_f = \sqrt{\frac{2KE}{m}} \quad v_f = \sqrt{\frac{2 \cdot 1000}{15}} = \sqrt{133.3} \rightarrow \boxed{36.5 \text{ m/s}}$

D.) $v_f = 36.5 \text{ m/s} \quad \theta = 45^\circ \quad g = 9.81 \text{ m/s}^2 \quad \sin 90 = 1$

$R = \frac{(36.5)^2 (1)}{9.81} = \frac{1332.25}{9.81} = \boxed{136 \text{ m}}$

2.3 a.) Force of friction $(F_f) = \mu_k \cdot m \cdot g = (0.05) (40) (9.81) = 19.6 \text{ N}$

$W_f = (19.6 \text{ N}) (40 \text{ m}) = \boxed{784 \text{ J}}$

b.) $P = \frac{W}{t} = \frac{784 \text{ J}}{30 \text{ s}} = \boxed{26.13 \text{ W}}$

2.4 $(0.003 \text{ kW}) (8760 \text{ hrs}) = 26.28 \text{ kWh}$

$\boxed{D = 2.33}$

Unit 6

3.1 $m_1 = m_2 = 20 \times 10^{-25} \text{ kg}$, $v_1 = 350 \text{ m/s}$, $v_2 = -350 \text{ m/s}$

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

$$v_f = 0$$

$$C = 0 \text{ ms}^{-1}$$

3.2 $P_{\text{total}} = \sqrt{p_1^2 + p_2^2 + 2 \cdot p_1 \cdot p_2 \cdot \cos(45^\circ)}$

$$P_{\text{total}} = \sqrt{(7000 \times 10^{-25} \text{ kg m/s})^2 + (7000 \times 10^{-25} \text{ kg m/s})^2 + 2 \cdot (7000 \times 10^{-25} \text{ kg m/s})^2 \cdot \cos(45^\circ)}$$

$$\rightarrow \sqrt{49 \times 10^{-40} + 49 \times 10^{-40} + 2 \cdot 49 \times 10^{-40} \cdot \frac{\sqrt{2}}{2}}$$

$$= 12935 \times 10^{-23}$$

$$v_f = \frac{P_{\text{total}}}{m_{\text{total}}} = \frac{12935 \times 10^{-23}}{40 \times 10^{-23}} = \boxed{323.4 \text{ m/s}}$$

3.3 a.) $J = F \cdot t = 4000 \text{ N} \cdot 0.20 \text{ s} = \boxed{800 \text{ N/s}}$

b.) $J = \Delta p = m \Delta v$ $J = -800 \text{ N/s}$, $m = 200 \text{ kg}$, $v_i = 2.8 \text{ m/s}$

$$\Delta v = v_f - v_i$$

$$v_f = \frac{J}{m} + v_i$$

$$v_f = \frac{-800}{200} + 2.80$$

$$= \boxed{-1.20 \text{ m/s}}$$

3.4 $C = \text{Elastic}$

3.5 a.) $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$

$$((30,000 \text{ kg})(10.850 \text{ m/s})) + ((110,000 \text{ kg})(0)) = 30,000 + 110,000 \cdot v_f$$

$$25,500 = 140,000 \cdot v_f$$

$$v_f = \frac{25500}{140,000} = \boxed{.182 \text{ m/s}}$$

b.) $KE = \frac{1}{2} (30,000 \text{ kg}) (10.850)^2 + \frac{1}{2} (110,000) \cdot 0^2$

$$= 10,837.5 \text{ J}$$

$$KE_f = \frac{1}{2} (140,000 \cdot 0.182^2) = 2,318.7 \text{ J}$$

kinetic energy lost =

$$\Delta KE = 10,837.5 - 2,318.7 = \boxed{8518.8 \text{ J}}$$

Unit 7

4.1 $T_b = 3 (40 \text{ kg}) (9.81) = 3.392 = 1176 \text{ N/m}$

$$T_c = 2 (60 \text{ kg}) (9.81) = 2.588 = 1176 \text{ N}$$

$$\boxed{A = \text{remain motionless}}$$

$$4.2 \quad \tau_w = 196 \text{ N} \cdot 1.00 \text{ m} = 196 \text{ N/m}$$

$$F_2 = \frac{196 \text{ N}}{.50} = 392 \text{ N}$$

$$F_2 = F_1 + \tau_w = 392 = F_1 + 196$$

$$F_1 = 196 \text{ N}$$

Force $F_1 = 196 \text{ N}$	↓
Force $F_2 = 392 \text{ N}$	↑ .5m

Unit 8

$$5.1 \quad a.) \quad I = \frac{1}{2} m r^2 = \frac{1}{2} (1 \text{ kg}) (0.008 \text{ m})^2$$

$$\text{moment of inertia} = 0.0032 \text{ kg/m}^2$$

$$b.) \quad I = \tau \cdot a \rightarrow 0.0032 (2)$$

$$\tau = 0.0064 \text{ N/m}$$

$$5.2 \quad a.) \quad \vec{r} = -0.20\hat{i} + 0.20\hat{j}$$

$$\vec{r} = \hat{i}(0.20 \cdot 0 - 0 \cdot 0) + \hat{j}(-20 \cdot 0 - 12 \cdot 0) + \hat{k}(-0.20 \cdot 0 - 12 \cdot 0.20)$$

$$\vec{r} = -2.4\hat{k} \text{ Nm}$$

$$b.) \quad \vec{r} = \hat{i}(0.20 \cdot 0 - F_y \cdot 0) + \hat{j}(-0.20 \cdot 0 - F_2 \cdot 0) + \hat{k}(-0.20 \cdot F_y - 0.20 \cdot F_x)$$

$$\vec{r} = \hat{k}(-0.20 F_y - 0.20 F_x)$$

$$F_y + F_x = -12$$

$$F_y = -12 \text{ N}$$

$$\vec{F} = 0\hat{i} - 12\hat{j} \text{ N}$$